

Air Handling Unit Specifications

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Project: Lakefront Office Tower — Phase II

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SECTION 23 73 13 — CENTRAL-STATION AIR HANDLING UNITS

1.0 SCOPE OF WORK

Furnish and install factory-assembled central-station AHUs per drawings M2.01–M2.04. Units serve office floors 3–18, the data center anteroom, and the rooftop mechanical penthouse.

2.0 PERFORMANCE SCHEDULE

Tag	Service	CFM	ESP (in.w.g.)	Cooling (MBH)	Heating (MBH)	OA %
AHU-1	Low Floors 3–9	28,500	3.50	320	180	25%
AHU-2	High Floors 10–18	26,000	3.75	295	165	25%
AHU-3	Data Anteroom	8,400	2.25	120	60	15%
AHU-4	Penthouse MER	3,200	1.50	48	24	100%

3.0 EQUIPMENT SPECIFICATIONS

3.1 Cabinet Construction

Double-wall panel, min 2-in. insulation (R-13), 20-ga G90 galvanized outer skin, 22-ga painted inner liner. Thermal break at all joints. Cabinet leakage $\leq 1\%$ at 1-in. w.g. per ASHRAE 111.

3.2 Supply Fan Assembly

Parameter	Requirement
Fan Type	Airfoil backward-inclined centrifugal
Drive	VFD — NEMA Premium IE3, 460V/3ph/60Hz
Bearing Life	L-10 $\geq 200,000$ hrs (ABMA Std 9)
Vibration	≤ 0.10 in/sec RMS (AMCA 204)
Isolation	2-in. deflection spring + neoprene pad

3.3 Coils & Filtration

Component	Specification	Standard
Cooling Coil	6-row CHW, copper 5/8-in. OD, Al fins 12–14 FPI, max 500 FPM face velocity	AHRI 410
Heating Coil	2-row HHW, copper tubes, Al fins, EWT 140°F / LWT 120°F	AHRI 410
Pre-Filter	2-in. pleated, MERV 8, ΔP 0.10/0.50 in. initial/final	ASHRAE 52.2
Final Filter	4-in. rigid bag, MERV 13, ΔP 0.25/1.00 in. initial/final	ASHRAE 52.2

4.0 SCHEMATIC — TYPICAL AHU ASSEMBLY

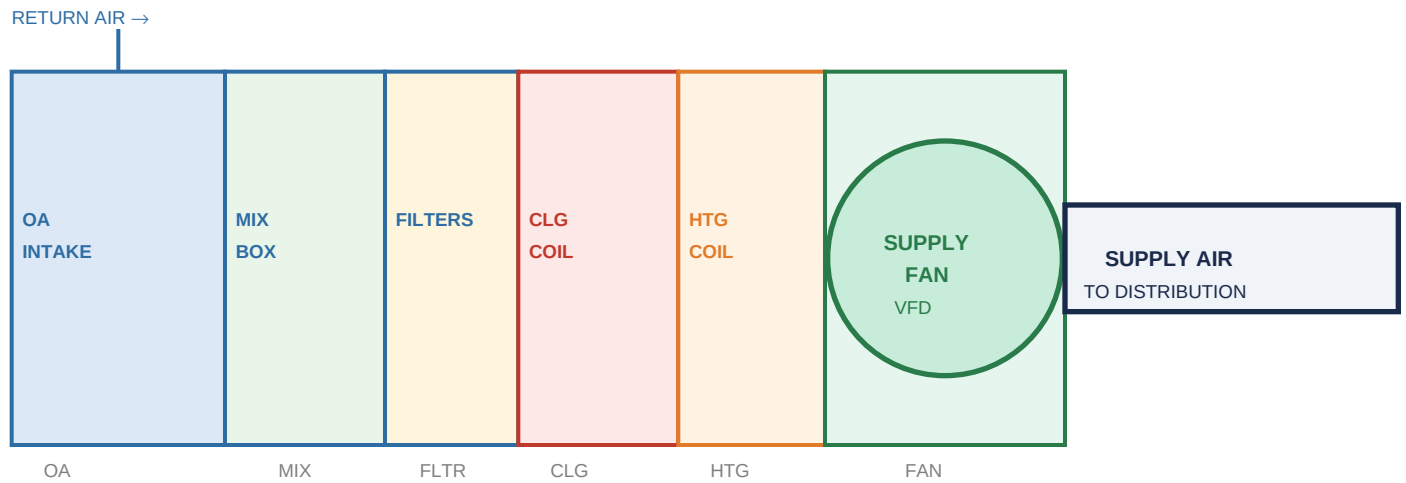


Figure 1 — Typical AHU Section Assembly (Not to Scale)

5.0 ACCEPTABLE MANUFACTURERS

Manufacturer	Model Series	Notes
Daikin Applied	Rebel / Vision	Basis of Design
Carrier	39M AquaForce	Approved Equal
Trane	Performance Climate Changer	Approved Equal
Johnson Controls	YVAA	Approved Equal

END OF SECTION 23 73 13